CONDITIONS NECESSARY AND SUFFICIENT FOR THE HEREDITARY LAWS OF LARGE NUMBERS

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Suppose the real-valued functions $f_1, f_2, \ldots$ are independent and have the same distribution, on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

We have two fundamental results:

STRONG LAW of LARGE NUMBERS: If $\mathbb{E}(|f_1|) < \infty$, then

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} f_n = \mathbb{E}(f_1), \quad \mathbb{P}-a.e.$$

And conversely: if

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} f_n = 0, \quad \mathbb{P}-a.e.,$$

then $\mathbb{E}(|f_1|) < \infty$, and $\mathbb{E}(f_1) = 0$.

WEAK LAW of LARGE NUMBERS: If $\lim_{N\to\infty} \big(N \cdot \mathbb{P}(|f_1| > N)\big) = 0$, then

$$\lim_{N\to\infty} \left(\frac{1}{N} \sum_{n=1}^{N} f_n - \mathbb{E}\left(f_1 \cdot \mathbb{1}_{\{|f_1| \le N\}}\right) \right) = 0, \text{ in } \mathbb{P}\text{-probability}.$$

And conversely, if there exist real numbers $D_1, D_2, \ldots$ such that

$$\lim_{N\to\infty}\left(\frac{1}{N}\sum_{n=1}^{N} f_n - D_N\right) = 0 \text{ in } \mathbb{P}\text{-probability}, \text{ then}$$

$$\lim_{N\to\infty}\left(N \cdot \mathbb{P}(|f_1|>N)\right) = 0.$$

These results have the following "hereditary" versions:

<u>KOMLÓS (1967) H-SLLN</u>: Suppose the measurable functions $f_1, f_2, \ldots$ satisfy $\sup_{n\in\mathbb{N}} \mathbb{E}(|f_n|) < \infty$

AND NOTHING ELSE.

There exist then a measurable $f_*: \Omega \to \mathbb{R}$ and a subsequence $f_{k_1}, f_{k_2}, \ldots$ of $f_1, f_2, \ldots$ along which, and along whose every subsequence,

$$\lim_{N\to\infty} \frac{1}{N}\sum_{n=1}^{N} f_{k_n} = f_*, \quad \mathbb{P}\text{-a.e.}$$

Furthermore, $\mathbb{E}(|f_*|) < \infty$.

This limiting f_* is a "randomized expectation" in the following sense. As A. RÉNYI showed, every bounded in $\mathbb{L}^0$ sequence $f_1, f_2, \ldots$ contains a so-called <u>determining subsequence</u> $f_{k_1}, f_{k_2}, \ldots$ with the stable convergence property

$$\lim_{n\to\infty} \mathbb{P}\left(f_{k_n} \le x, B\right) = \int_B \mathbf{H}(x,\omega)\, \mathbb{P}(d\omega), \quad \forall\, B \in \mathcal{F}$$

valid for every $x \in \mathbb{D}$, a countable and dense subset of $\mathbb{R}$.

Here $\mathbb{R} \times \Omega \ni (x,\omega) \mapsto \mathbf{H}(x,\omega) \in [0,1]$ is a random probability distribution function, with induced probability measure μ_ω on $\mathcal{B}(\mathbb{R})$, and measurable with respect to the tail σ-algebra

$$\mathcal{T} = \bigcap_{n \in \mathbb{N}} \mathcal{T}_n, \qquad \mathcal{T}_n \triangleq \sigma\left(f_{k_n}, f_{k_{n+1}}, \ldots\right).$$

In terms of them, the limiting f_* can be expressed as a generalized mean (expectation)* :

$$f_* = \int_{\mathbb{R}} x\, d\mathbf{H}(x,\omega) = \int_{\mathbb{R}} x\, \mu_\omega(dx).$$

<u>K-SCHACH. (2023) H-WLLN</u>: Suppose the functions $f_1, f_2, \ldots$ satisfy

(*) $$\lim_{M\to\infty}\Big(M \cdot \sup_{n\in\mathbb{N}} \mathbb{P}\big(|f_n| > M\big)\Big) = 0.$$

There exist then a subsequence $f_{k_1}, f_{k_2}, \ldots$ and $\mathcal{F}$-measurable $D_1, D_2, \ldots$ satisfying

$$\mathbb{P}\big(|D_N| \le N,\ \forall N\in\mathbb{N}\big) = 1,$$

$$\lim_{n\to\infty} \mathbb{E}\Big(f_{k_n} \cdot \mathbb{1}_{\{|f_{k_n}|\le N\}} \cdot \xi\Big) = \mathbb{E}\big(D_N \cdot \xi\big), \quad \forall\, \xi \in \mathbb{L}^2$$

as well as

$$\lim_{N\to\infty}\left(\frac{1}{N}\sum_{n=1}^{N} f_{k_n} - D_N\right) = 0 \ , \quad \text{in } \mathbb{P}\text{-probability}$$

along this, and along every further, subsequence.

<u>Important Points</u>:

1) There exist sequences that satisfy (*) as well as

$$\mathbb{E}(|f_n|) < \infty,\ \forall\, n\in\mathbb{N} \ ; \qquad \lim_{n\to\infty} \mathbb{E}(|f_n|) < \infty$$

(i.e., contain no subsequence satisfying the KOMLÓS Condition).

2) Neither (*), nor the KOMLÓS Condition, are necessary in their respective contexts.

The conditions in these two results are sufficient, but not necessary.

THEOREM: N&S in the H-SLLN

Consider real-valued, measurable $f_1, f_2, \dots$ on a given probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

(i) Suppose that

(1) $$\sup_{n \in \mathbb{N}} \mathbb{E}^{\mathbb{P}}\left(|f_{k_n}| \cdot \mathbb{1}_{A_j} \right) < \infty, \quad \forall j \in \mathbb{N}$$

holds for some subsequence $f_{k_1}, f_{k_2}, \dots$ and sets $A_1 \subseteq A_2 \subseteq \dots \subseteq A_j \subseteq \dots$ with $\lim_{j\to\infty} \mathbb{P}(A_j) = 1$.

There exist then a measurable $f_*: \Omega \to \mathbb{R}$ and a (further, relabelled) subsequence along which, and along whose every further subsequence, we have

(2) $$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} f_{k_n} = f_*, \quad \mathbb{P}\text{-a.e.}$$

(ii) Conversely, suppose that (2) holds along some subsequence $f_{k_1}, f_{k_2}, \dots$ and along all its subsequences, for some f_* measurable. Then there exist sets $A_1 \subseteq A_2 \subseteq \dots$ with $\lim_{j\to\infty} \mathbb{P}(A_j) = 1$, such that (1) holds.

(iii) The validity of (1) as stated, is equivalent to the existence of a probability measure $\mathbb{Q} \sim \mathbb{P}$ on $(\Omega, \mathcal{F})$, such that

(3) $$\sup_{n \in \mathbb{N}} \mathbb{E}^{\mathbb{Q}}\left(|f_{k_n}| \right) < \infty.$$

(iv) Under any of the equivalent conditions (1) or (3), the limit function f_* of (2) is integrable, i.e. $\mathbb{E}^{\mathbb{P}}(|f_*|)<\infty$ if, and only if,

(4) $$\sup_{n\in\mathbb{N}} \mathbb{E}^{\mathbb{P}}\left(|f_{k_n}|\cdot \mathbb{1}_{A_n}\right)<\infty$$

holds for some subsequence $f_{k_1}, f_{k_2}, \dots$, and some sequence $A_1\subseteq A_2\subseteq\dots$ of sets in $\mathcal{F}$ with $\lim_{n\to\infty}\uparrow \mathbb{P}(A_n)=1$.

THEOREM: N&S in the H-WLLN

Consider real-valued, measurable $f_1, f_2, \dots$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

(i) Suppose that for some subsequence $f_{k_1}, f_{k_2}, \dots$ and sets $A_1 \subseteq A_2 \subseteq \dots$ with $\lim_{j\to\infty} \mathbb{P}(A_j) = 1$, we have

(5) $$\lim_{M\to\infty} \Big(M \cdot \sup_{n\in\mathbb{N}} \mathbb{P}\big(|f_{k_n}| \cdot \mathbb{1}_{A_j} \big) \Big) = 0, \quad \forall j \in \mathbb{N}.$$

There exist then $\mathcal{F}$-measurable "correctors" $D_1, D_2, \dots$ such that

(6) $$\lim_{N\to\infty} \Big(\frac{1}{N} \sum_{n=1}^{N} f_{k_n} - D_N \Big) = 0, \quad \text{in } \mathbb{P}\text{-probability}$$

holds along the subsequence and along all its subsequences.

Each corrector D_N takes values in $[-N, N]$; each $D_N \mathbb{1}_{A_j}$ is the weak $\mathbb{L}^2$-limit of $\big(f_{k_n} \mathbb{1}_{A_j \cap \{|f_{k_n}| \le N\}} \big)_{n\in\mathbb{N}}$, and we have $\frac{1}{N} \mathbb{E}(D_N^2 \mathbb{1}_{A_j}) \xrightarrow[N\to\infty]{} 0, \quad \forall j \in \mathbb{N}.$

(ii) Conversely, if (6) holds as in (i), there exist a subsequence $f_{k_1}, f_{k_2}, \dots$ and sets $A_1 \subseteq A_2 \subseteq \dots$ in $\mathcal{F}$ with $\lim_{j\to\infty} \mathbb{P}(A_j) = 1$, such that (5) holds.

(iii) The validity of (4) as in (i), is equivalent to the existence of a probability measure $Q \sim P$ on $\mathcal{F}$, such that for some subsequence we have

$$\lim_{M\to\infty}\Big(M \cdot \sup_{n\in\mathbb{N}} Q\big(|f_{k_n}|>M\big)\Big)=0. \tag{7}$$